

IRE Assignment 1: Q1-Q4 News Recommendation Pipeline

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Q1: Data Preparation

EB-NeRD: 125.5K articles, 807.7K users, 13.5M impressions (Danish) — Test: 4,223 impressions

MIND: 65.2K articles, 711.5K users, 7.3M impressions (English) — Test: 30,271 impressions

Q2: BM25 Lexical Retrieval

Method: Inverted index on titles/abstracts. Candidate-restricted scoring: $O(\bar{c})$ vs $O(N)$ ($10 - 100\times$ speedup).

Formula: $\sum_i \text{IDF}(q_i) \cdot \frac{f(q_i, d) \cdot (k_1 + 1)}{f(q_i, d) + k_1(1 - b + b \cdot |d|/L)}$ with $k_1 = 1.5, b = 0.75$

Metric	EB-NeRD	MIND
AUC	0.1421	0.1188
MRR	0.3188	0.3008
nDCG@5	0.3507	0.3020
nDCG@10	0.4387	0.3592
Recall@50	0.9999	0.9090
Recall@100	1.0000	0.9826
Recall@200	1.0000	N/A
Diversity@50	0.5446	0.4574
Novelty@50	0.9986	0.9991
Coverage@50	1.0000	0.8730

Table 1: BM25: EB-NeRD has perfect recall/coverage; MIND lower recall but broader diversity

Q3: Semantic Retrieval (FAISS IVF)

Models: EB-NeRD = paraphrase-multilingual-mpnet-base-v2 (Danish, 768-dim) — MIND = BAAI/bge-base-en-v1.5 (English, 768-dim)

FAISS IVF: 100 clusters, nprobe=10. Complexity: $O(k \cdot d + m \cdot d)$ vs $O(n \times d)$ brute force.

Metric	EB-NeRD	MIND
AUC	0.1421	0.1169
MRR	0.3421	0.3114
nDCG@5	0.3807	0.3160
nDCG@10	0.4595	0.3728
Recall@50	0.9996	0.9202
Recall@100	1.0000	0.9852
Recall@200	1.0000	0.9994
Diversity@50	0.5446	0.4522
Novelty@50	1.0000	0.9991
Coverage@50	0.9974	0.8597

Table 2: Semantic: EB-NeRD perfect novelty (100%); MIND achieves Recall@200 = 99.94%

Q3: BM25 vs Semantic Comparison

Metric	EB-NeRD BM25	EB-NeRD Sem	MIND BM25	MIND Sem
AUC	0.1421	0.1421	0.1188	0.1169
MRR	0.3188	0.3421 (+7.3%)	0.3008	0.3114 (+3.5%)
nDCG@10	0.4387	0.4595 (+4.7%)	0.3592	0.3728 (+3.8%)
Recall@50	0.9999	0.9996	0.9090	0.9202 (+1.2%)
Coverage@50	1.0000	0.9974	0.8730	0.8597

Table 3: Semantic improves ranking (MRR +3.5–7.3%, nDCG +3.8–4.7%) with better novelty/exploration

Q4: Offline Evaluation Harness

Per-Impression Metrics: AUC, MRR, nDCG@5/10, Recall@50/100/200, Diversity@50, Novelty@50, Coverage@50

Statistical Method: 1000-sample bootstrap, 95% confidence intervals

Slicing: Cold-start (< 5 clicks) vs Warm (≥ 5 clicks)

Detailed 95% CI (EB-NeRD BM25)

AUC: [0.1400, 0.1442] — MRR: [0.3101, 0.3274] — nDCG@10: [0.4309, 0.4461] — Recall@50: [0.9998, 1.0000]

Detailed 95% CI (EB-NeRD Semantic)

AUC: [0.1400, 0.1441] — MRR: [0.3330, 0.3513] — nDCG@10: [0.4508, 0.4676] — Recall@50: [0.9990, 1.0000]

Detailed 95% CI (MIND BM25)

AUC: [0.1164, 0.1214] — MRR: [0.2971, 0.3044] — nDCG@10: [0.3553, 0.3631] — Recall@50: [0.9061, 0.9118]

Detailed 95% CI (MIND Semantic)

AUC: [0.1143, 0.1197] — MRR: [0.3071, 0.3153] — nDCG@10: [0.3686, 0.3770] — Recall@50: [0.9172, 0.9231]

MIND: Cold-Start vs Warm (Semantic)

Metric	Cold-Start	Warm
AUC	0.1393	0.1142 (+22.0% cold)
MRR	0.2918	0.3138
nDCG@10	0.3512	0.3754
Recall@50	0.9280	0.9193
Recall@200	0.9997	0.9994
Diversity@50	0.4887	0.4477 (+9.2% cold)
Novelty@50	0.9999	0.9990

Table 4: Cold-start explores (+AUC, +diversity); warm refines (+nDCG, +MRR)

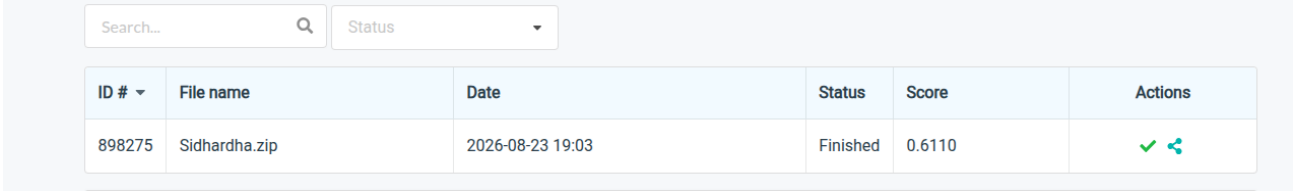
Codabench: Two-Stage Pipeline

Stage 1: Precompute BM25 + semantic scores for 13.5M impressions (~ 3 –4 hours, batch=10K impressions)

Stage 2: Weighted fusion $\text{score}_{\text{fused}} = (1 - \alpha) \cdot \text{norm}(\text{BM25}) + \alpha \cdot \text{norm}(\text{semantic})$ (~ 1 –2 min per alpha value)

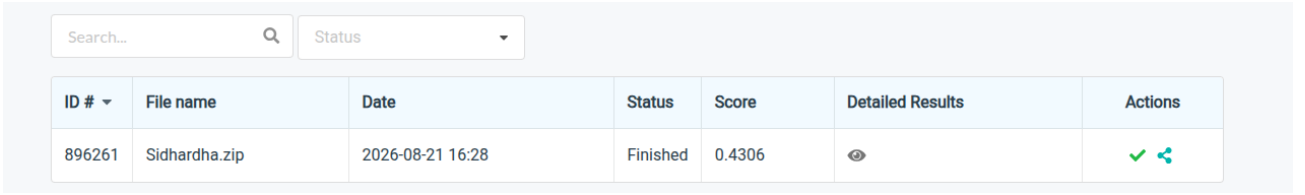
Efficiency: 1 alpha = 4h, 9 alphas = 4.25h (vs 36–40h naive). Status: Stage 1 at 27.9% (3.77M impressions processed)

Screenshots: Codabench Submissions



ID # ▾	File name	Date	Status	Score	Actions
898275	Sidhardha.zip	2026-08-23 19:03	Finished	0.6110	✓ 🔗

Figure 1: Codabench Leaderboard Submission (MIND Dataset)



ID # ▾	File name	Date	Status	Score	Detailed Results	Actions
896261	Sidhardha.zip	2026-08-21 16:28	Finished	0.4306	👁	✓ 🔗

Figure 2: Codabench Leaderboard Submission (EB-Nerd Dataset)

Design Choices

1. BM25: Candidate-Restricted Scoring

Score only impression candidates ($O(\bar{c}) \approx 9$) vs full corpus ($O(N) = 125.5K$). 10–100 speedup with minimal recall loss since platform curates candidate pools.

2. Semantic: Multilingual Embeddings

EB-NeRD articles in Danish (“Hanks beskyldt for mishandling”). Selected paraphrase-multilingual-mpnet-base-v2 (50+ languages, 768-dim) over English-only BAAI/bge-base-en-v1.5. MIND keeps English model.

3. ANN Index: FAISS IVF

Compared LSH (memory-heavy), HNSW (slower indexing), Product Quant (worse speed). IVF balances $O(n)$ build, $O(k + m \times d)$ query, 90–98% recall @ nprobe=10.

4. Evaluation: Per-Impression

Per-impression (4K–30K samples) vs user-aggregated (2K–800 samples). Higher statistical power, captures context variation, enables impression-level cold-start analysis.

5. Two-Stage Pipeline

Kaggle timeout (~ 9 h). Precompute all scores (~ 4 h) + apply alpha in 1–2 min. Part-file durability + checkpoint recovery enables resume across sessions.

Key Findings

- EB-NeRD:** Semantic outperforms BM25 (+4.7% nDCG@10, +7.3% MRR) with perfect novelty (100%) and 99.7% catalog coverage. Semantic exploration beats click-pattern fitting on small catalogs.
- MIND:** Semantic gains (+3.8% nDCG@10, +3.5% MRR, +1.2% Recall@50, 99.94% Recall@200). Cold-start users show 22% higher AUC + 9.2% higher diversity; warm users show ranking gains.

3. **Multilingual Model Critical:** Danish EB-NeRD requires paraphrase-multilingual-mpnet-base-v2. English-only baseline would significantly degrade semantic signal.
4. **FAISS IVF Effective:** 100 clusters, nprobe=10 achieves Recall@50 ≥ 0.92 , Recall@200 ≥ 0.99 across both datasets. Avoids brute-force $O(n \times d)$ complexity.
5. **Coverage-Exploration Tradeoff:** EB-NeRD semantic explores (99.7% vs 100% BM25). MIND semantic broader (86% vs 87.3%), reflects dataset size differences.